BLEKINGE TEKNISKA HÖGSKOLA

BTH

Interpretable Predictive Maintenance for SCANIA Trucks

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Introduction

- Logistics and supply chain – backbone of the economy.
- Vehicle reliability is critical. Unexpected breakdowns lead to:
  - Costly emergency repairs.
  - Supply chain delays.

Introduction (2)

- Traditional: Reactive Maintenance (Fix it when it breaks).
- Modern: Predictive Maintenance (Fix it before it breaks).
- **Key Challenge**: Current methods use Deep Learning (LSTMs/Transformers).
- Issue: “Black Boxes” → Low trust among domain experts.

Problem Statement

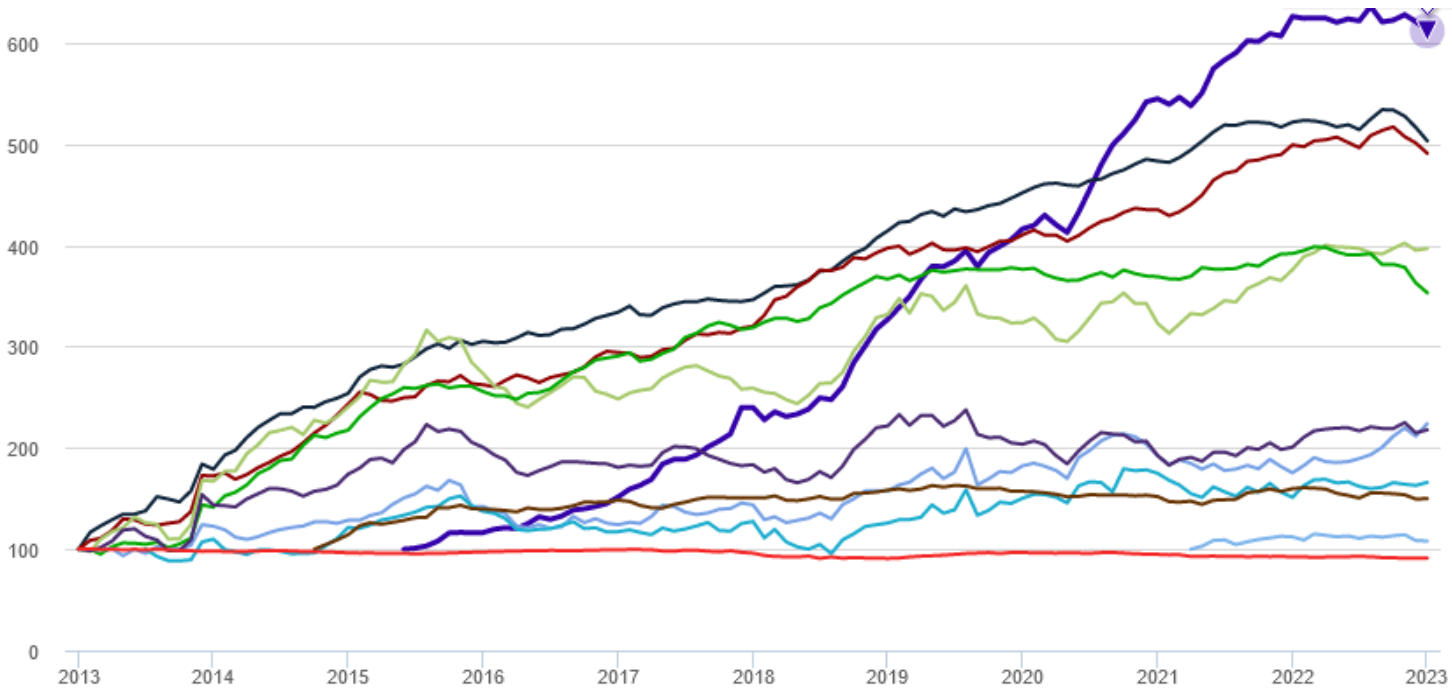
- Develop an interpretable mechanism to mine predictive failure patterns in trucks, and the predictions must be justifiable.

Solution

- Train an interpretable ML model to predict the failure of a truck.
- Justify the predictions.
- CRISP-DM methodology.

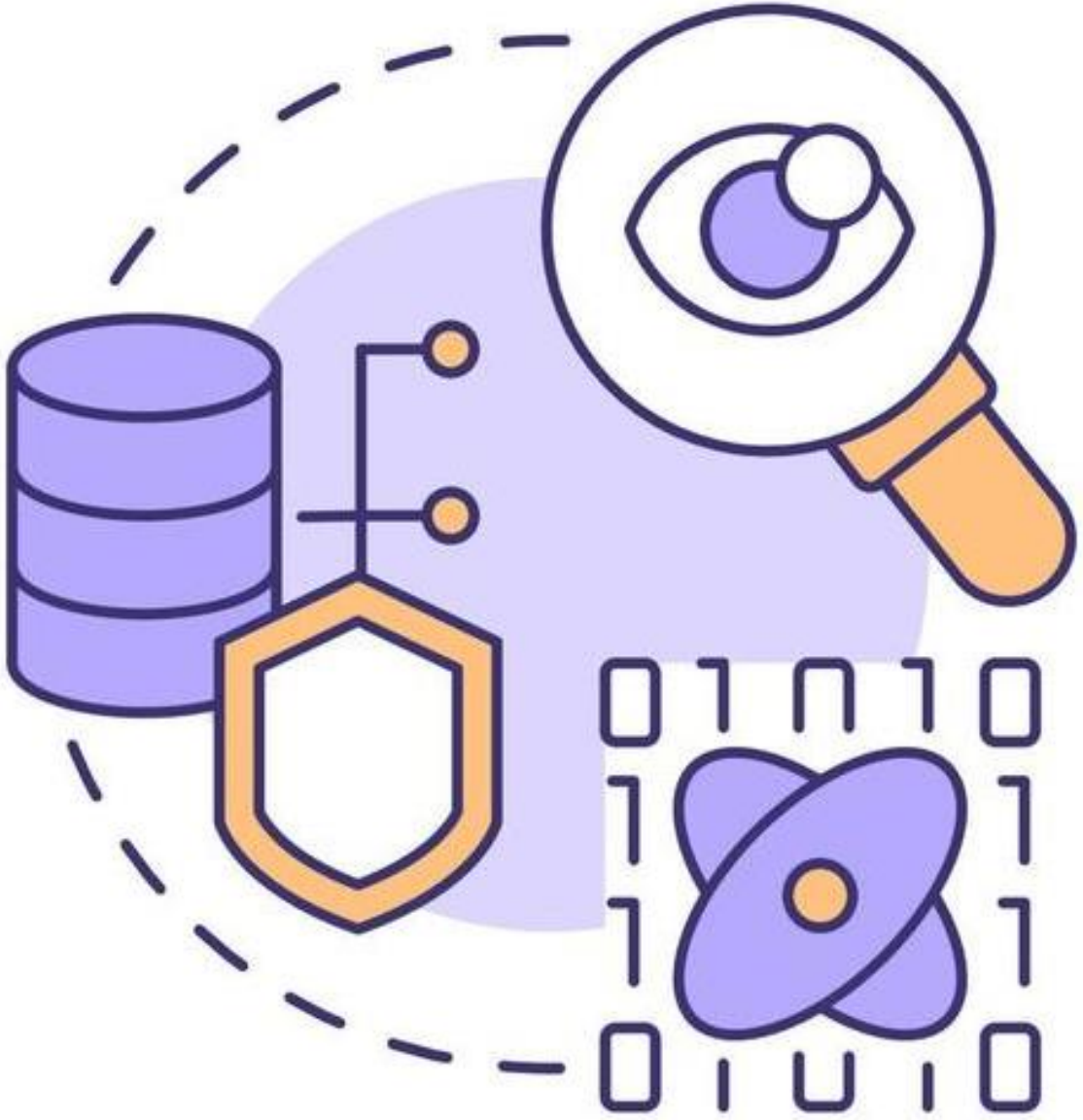
Dataset

- SCANIA Component X Dataset - Multivariate Time Series Data – 100+
- Imbalanced data
- Cumulative counters, histogram sensor readings.
- Feature engineering - preserve temporal information



Performance Metrics

- Metrics: Recall (Sensitivity)
  - Failure is a rare event.
- Explainability: Validate model trust using techniques like SHAP.



Questions & Discussion

Thank You